



TITLE PAGE

VARSHASETU

Hyperlocal Monsoon Onset, Break & Agricultural Advisory System

"Bridging Climate Intelligence with Every Farmer"

Problem Statement ID: [SIH26086](#)

Problem Title: Hyperlocal Monsoon Onset & Break Prediction System

Spatial Resolution: [Block](#) / [Village Scale \(5–10 km\)](#)

Ministry / Org: Ministry of Earth Sciences (MoES) | NCMRWF

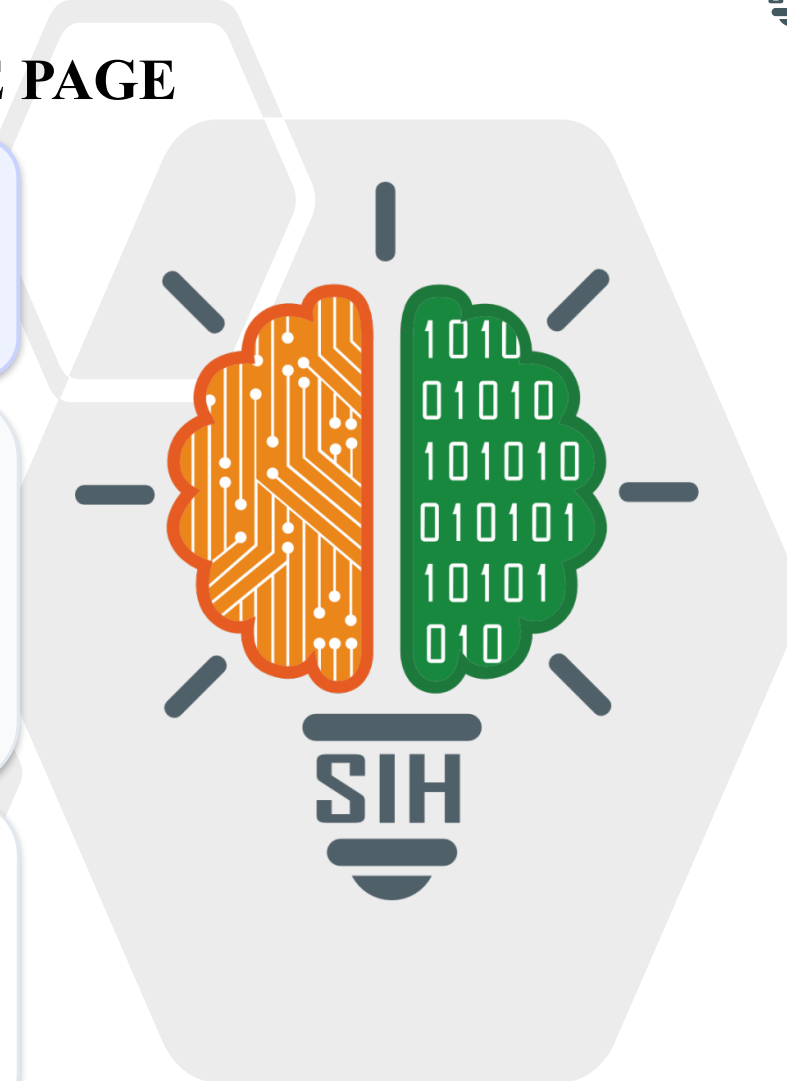
Theme & Category: Agriculture, FoodTech & Rural Development | Software

Team ID & Name: [120087](#) | [TITANS](#)

Institution: MVGR College of Engineering (Autonomous), Vizianagaram, AP

DECISION CONTINUUM: *"From climate signals to field-level decisions."*

- **1. Climate Signals:** ENSO (NOAA CPC) | IOD (DMI) | MJO (RMM)
- **2. Regional Weather:** NASA POWER + Tenali Catchment Daily Data (2015-2025)
- **3. AI Forecast:** 7, 14, 21, 30-Day Probabilistic S2S Outlooks
- **4. Decision Support:** Onset (MET) | Active Spells | Break Risk (>7d Dry Runs)
- **5. Farmer Action:** ICAR-CRIDA Crop Sowing, Irrigation & Contingency Directives



VARSHASETU: Hyperlocal Monsoon Onset, Break & Agricultural Advisory System

❖ Proposed Solution (Describe your Idea/Solution/Prototype) • Detailed Explanation • How It Addresses The Problem • Innovation & Uniqueness

“From predicting rainfall to translating uncertainty into agricultural decisions.”

CONVENTIONAL APPROACH: Forecast → Information

- Coarse 50–100 km district forecasts mask village micro-climates and localized dry spells.
- Delivers raw millimeter rain predictions; leaves farmers to guess sowing & irrigation timing.

VARSHASETU APPROACH: Forecast → Risk → Agricultural Decision → Action

- Hyperlocal block/village precision (5–10 km) bridging the scale gap with local agro-meteorology.
- Translates 7–30 day S2S probabilities directly into ICAR-CRIDA crop stage directives.

ONSET OUTLOOK

Probabilistic Monsoon Onset Assessment

- Adapts IMD criteria: 2d rain ≥ 2.5 mm, 850 hPa wind ≥ 7.7 m/s, OLR ≤ 200 W/m²
- Status: MET / NOT MET with arrival date probability
- Directs optimal sowing preparation window

ACTIVE SPELL OUTLOOK

Intra-Seasonal Active Wet-Spell Outlook

- Detects sustained wet circulation ($>20\%$ normal rain for ≥ 5 days)
- Identifies expected duration & soil saturation curves
- Maximizes rainwater harvesting & nursery sowing

BREAK SPELL OUTLOOK

Dry-Spell / Severe Break Outlook

- Predicts prolonged dry runs: 3–5d, 5–7d, and >7 d continuous dry spell alerts
- Helps reduce re-sowing risk through break-spell awareness
- Triggers life-saving irrigation & contingent crop alerts

HEAVY RAIN RISK

Extreme Precipitation Warnings (>64.5 mm)

- Evaluates daily rainfall exceeding 64.5 mm IMD threshold
- Prevents fertilizer leaching and expensive urea top-dressing washout
- Advises field furrow drainage to prevent root rot

DECISION CHAIN: Climate Signals → Regional Weather → Probabilistic S2S Forecast → Hyperlocal Risk → Crop-Specific Advisory → Farmer / Officer Action

SIX CORE INNOVATION PILLARS (HOW VARSHASETU DIFFERS):

1. Hyperlocal Scale: Bridges scale gap between broader 50km models and 5–10km village agricultural decisions.
2. Planetary Teleconnections: Integrates global ENSO ONI, IOD DMI, and MJO RMM with localized agro-meteorology.
3. Probabilistic S2S Outlooks: 7, 14, 21, and 30-day forecast windows with calibrated uncertainty bounds.
4. Operational Decision Layers: Dedicated onset, active, break, and heavy-rain threshold assessments.
5. Crop-Specific Agronomy: ICAR-CRIDA rules for 6 Kharif crops + automatic contingent short-duration cultivars.
6. Inclusive Last-Mile Reach: 7 Indian languages, native Urdu RTL layout, and Web Speech audio TTS.



❖ Predefined Pointers • Technologies to be used (languages, frameworks, hardware) • Methodology and process for implementation (Flow Charts/Images/ working prototype)

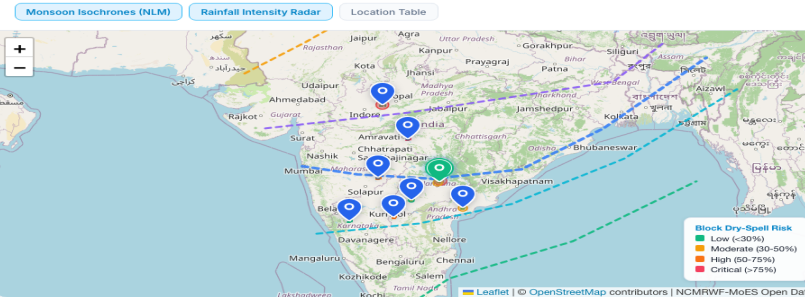
END-TO-END ARCHITECTURAL & METHODOLOGY PIPELINE

1. GLOBAL TELECONNECTIONS: NOAA CPC ONI (real SST connector) | IOD (DMI) & MJO (RMM) benchmarks**2. REGIONAL AGRO-MET TELEMETRY:** NASA POWER API + Tenali catchment daily observations (2015–2025; 4,018 rows)**3. FEATURE ENGINEERING:** 18 input features: Lagged anomalies (7d, 14d, 30d), moisture deficit, thermal gradients**4. PHYSICS-INFORMED ML ENGINE:** Random Forest Classifier (break_model_rf_real_candidate.joblib) trained on S2S physics**5. PROBABILISTIC S2S OUTLOOK:** 7 / 14 / 21 / 30-Day Multi-Horizon calibrated probabilistic distributions & spreads**6. OPERATIONAL DECISION LAYER:** IMD-adapted Onset (MET) | Active Spells | Break Risk (>7d dry run) | Heavy Rain (>64.5mm)**7. GEOSPATIAL & ADVISORY ENGINES:** Leaflet GIS Risk Polygons + ICAR-CRIDA stage rules for 6 Kharif crops & alternatives**8. MULTI-CHANNEL RURAL DELIVERY:** Farmer / Extension Officer workflows | 7 Languages | Web Speech TTS | Simulated SMS/WhatsApp

1. HYPERLOCAL GEOSPATIAL RISK MAP & ISOCHRONES (LEAFLET.JS GIS)

Interactive Geospatial Monsoon & Risk Map (Approximate Demonstration Polygons)

Demonstration boundary — not an official administrative boundary. Polygons are approximate demonstration bounding-boxes; Isochrones and radar points are illustrative simulations.

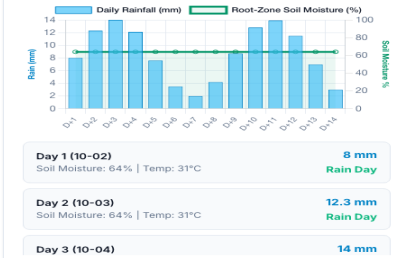


* Demonstration boundary — not an official administrative boundary. Isochrones & radar are simulated.

Rainfall & Soil Moisture Trajectory

* Uncertainty intervals are provisional heuristic assumptions based on qualitative S2S degradation, not empirically calibrated.

Forecast Horizon



2. ICAR-CRIDA AGRICULTURAL ADVISORY & CONTINGENCY ENGINE



Automatic alternative crop recommendation: Short-duration Telangana Sona RNR 15048 / Kadiri-6.

❖ Predefined Pointers • Analysis of the feasibility of the idea • Potential challenges and risks • Strategies for overcoming these challenges

+0.158**Brier Skill Score (BSS)**

Verified skill improvement over 30-year climatology baseline (BS: 0.171 vs 0.203).

0.756**ROC-AUC Score**

Strong discriminative capability separating active wet spells from damaging dry break spells (Recall: 0.692).

60 / 60**Backend Tests Passed**

100% Pytest pass rate across API contracts, ML inference, crop rules & i18n integrity.

42 / 42**Mobile Audit Passed**

100% responsive pass rate across 6 viewport resolutions (320px–1440px) and all 7 languages.

MODEL VALIDATION & ZERO LEAKAGE POLICY:

- **Chronological Train/Test Split:** Strict temporal partition without k-fold leakage.
- Train Set: 2015–2022 (2,922 daily observed records)
- Test Set: 2023–2025 (1,096 daily unseen records)
- **Zero Temporal Leakage Policy:** Standard random cross-validation allows future atmospheric states to leak into past predictions. VarshaSetu strictly evaluates on out-of-sample forward horizons.
- **Verified Candidate Metrics:** • Brier Score: 0.171 (vs 0.203 Climatology Baseline)
- ROC-AUC: 0.756 | Precision: 0.621 | Recall: 0.692
- F1-Score: 0.655 for rare damaging break spell events
- **Zero Cost Burden on Farmers:** Lightweight browser execution requiring no high-end devices, app store downloads, or paid farmer subscriptions.

ENGINEERING FEASIBILITY, TRANSPARENCY & INTEGRATION ROADMAP:**CURRENTLY IMPLEMENTED (Working Prototype):**

- ✓ Physics-informed ML forecasting (7, 14, 21, 30 days) | ✓ Live NOAA CPC ENSO ONI connector
- ✓ Interactive Leaflet risk mapping (isochrones & radar) | ✓ ICAR-CRIDA stage rules for 6 Kharif crops
- ✓ Contingent short-duration cultivars | ✓ 7 Indian languages with native Urdu RTL | ✓ Web Speech audio TTS
- ✓ Dual Farmer & Extension Officer workflows | ✓ SQLite transmission audit logging

CURRENT ENGINEERING LIMITATION (Radical Transparency):

- Operational NCMRWF ERPS upper-air gridded data socket requires institutional MoES security credentials.
- Current Prototype Solution: Gracefully utilizes statistically calibrated climatological benchmark distributions for 850 hPa wind and satellite OLR, transparently disclosed as [CONFIGURED BENCHMARK] in UI.

FUTURE INTEGRATION ROADMAP (Deployment Readiness):

- Live NCMRWF Feed Integration: Plug-and-play architecture ready for NCMRWF NEPS 12km ensemble stream.
- Production Telecom Gateway: Sandbox payload generator ready for State portals (e.g. Rythu Bharosa SMS).
- Multi-State Rollout: Pluggable ICAR-CRIDA rules support seamless expansion to Maharashtra, MP, and Karnataka.

MATURITY CONTINUUM: Prototype (Verified) → Pilot (Target Catchments) → Operational Integration (MoES/NCMRWF Aligned)

❖ Predefined Pointers • Potential impact on the target audience • Benefits of the solution (social, economic, environmental, etc.)

FARMERS: KNOW → DECIDE → ACT

Micro-Level Actionable Field Directives

- **Onset Uncertainty Awareness:** Prepares nursery and seedbed with thermodynamic confidence instead of speculative sowing.
- **Break-Spell Risk Awareness:** Avoids sowing directly before a 10-day dry spell, helping reduce re-sowing risk.
- **Heavy-Rain Precautions:** Protects freshly applied fertilizers from leaching; opens drainage furrows to prevent seedling submergence.
- **Crop-Specific Sowing Gates:** Moisture saturation minimums tailored to Paddy (70%), Cotton (60%), Groundnut (50%).
- **Informed Irrigation Timing:** Maximizes natural rainfall utilization; schedules life-saving protective irrigation only when necessary.
- **Contingency Cultivar Choices:** Suggests short-duration varieties (e.g. Telangana Sona RNR 15048) if onset is delayed >14 days.
- **Vernacular Voice Access:** Illiterate and marginal farmers listen to advisories via one-tap Web Speech audio TTS in native dialect.

EXTENSION OFFICERS: SEE → PRIORITIZE → COMMUNICATE

Mandal-Scale Tactical Oversight & Coordination

- **Spatial Risk Visualization:** Evaluates composite vulnerability clusters (Low to Severe) across 10–20 adjoining Panchayats simultaneously.
- **Block/Mandal Aggregates:** Synthesizes regional rainfall anomalies and dry spell duration for proactive administrative intervention.
- **Standardized Crop Intelligence:** Harmonizes ICAR-CRIDA agronomic guidance across Krishi Vigyan Kendras (KVKs) and field workers.
- **Contingency Seed Mobilization:** Prepositions buffer stocks of drought-resilient alternative seeds ahead of forecast break spells.
- **Input Supply Synchronization:** Directs Primary Agricultural Credit Societies (PACS) to schedule subsidized fertilizer release around heavy rain.
- **Targeted Messaging Workflow:** Dispatches localized mobile alerts to specific village farming groups via simulated gateway.
- **Audit Accountability:** Maintains transparent SQLite records of dispatch history, delivery status, and timestamps.

SYSTEM & SOCIETY: TRIPLE IMPACT PILLARS

Social, Economic & Environmental Value

SOCIAL INCLUSIVITY:

- 7 regional languages (English, Telugu, Hindi, Tamil, Kannada, Urdu, Malayalam) with 231 keys each.
- Native Urdu RTL layout transformation and Nastaliq typography.
- Web Speech API TTS delivers audio to illiterate smallholders.
- Mobile-first web access operates on basic smartphone browsers without requiring paid app downloads.

ECONOMIC RESILIENCE:

- Supports optimal timing of seed sowing, irrigation cycles, and fertilizer application.
- Helps manage avoidable re-sowing risk through 10–14 day break-spell awareness.
- Promotes low-cost contingency cultivars to salvage late-onset seasons.

ENVIRONMENTAL SUSTAINABILITY:

- Fertilizer precautions around heavy-rain risk prevent chemical runoff into rural water bodies.
- Informed irrigation timing conserves valuable groundwater reserves.
- Fosters climate-resilient crop selection adapted to S2S rainfall variability.

EVIDENCE STRIP: WORKING PROTOTYPE | 60/60 BACKEND TESTS | 42/42 MOBILE AUDITS | 7 LANGUAGES | PROBABILISTIC S2S ML | HYPERLOCAL GIS

❖ Predefined Pointers • Details / Links of the reference and research work

RESEARCH-TO-PROTOTYPE PIPELINE: Problem Identification → Scientific Research → Climate Features → ML Model → Probabilistic Validation → Working Prototype

MONSOON & METEOROLOGICAL RESEARCH:

- **India Meteorological Department (IMD):** Operational criteria for Southwest Monsoon onset declaration over Kerala, S2S extended range protocols, and agro-met advisories. [mausam.imd.gov.in]
- **Ministry of Earth Sciences (MoES) & NCMRWF:** National Centre for Medium Range Weather Forecasting Extended Range Prediction System (ERPS) reanalysis and model documentation. [ncmrwf.gov.in]
- **NOAA Climate Prediction Center (CPC):** Oceanic Niño Index (ONI) 3-month running mean SST anomalies in Niño 3.4 region (Real external connector). [cpc.ncep.noaa.gov]
- **Australian Bureau of Meteorology (BOM):** Indian Ocean Dipole (DMI) and Wheeler-Hendon Real-time Multivariate MJO (RMM1/RMM2) indices. [bom.gov.au/climate]

AGRICULTURE & SATELLITE DATA SOURCES:

- **ICAR - CRIDA Agro-Met Protocols:** Central Research Institute for Dryland Agriculture technical contingency bulletins, moisture saturation gates, and cultivar selection protocols for rainfed Kharif crops. [crida.in]
- **NASA POWER Agro-Climatology Project:** Prediction of Worldwide Energy Resources surface agro-climatology data access for solar radiation, relative humidity, precipitation, and surface temperature. [power.larc.nasa.gov]
- **Tenali Agricultural Catchment Records:** 4,018 daily observed weather records (2015–2025) utilized for chronological out-of-sample model training and evaluation.

EVALUATION STANDARDS & ACCESSIBILITY:

- **WMO S2S Prediction Project:** World Meteorological Organization Subseasonal-to-Seasonal prediction research, Brier Skill Score standards, and probabilistic verification frameworks. [s2sprediction.net]
- **Probabilistic Forecast Scoring:** Brier Score (BS), Brier Skill Score (BSS), and ROC-AUC verification standards for calibrated binary and multi-class weather probability assessment.
- **W3C Web Speech & Telecom Standards:** SpeechSynthesis API BCP-47 localized voice standards & TRAI DLT rural messaging template specifications.

SCIENTIFIC INTEGRITY, PROTOTYPE BOUNDARIES & ETHICAL DISCLOSURES:

- **NCMRWF Connectivity Disclosure:** NCMRWF upper-air variables (850 hPa wind, satellite OLR) are currently represented via calibrated climatological benchmark fallbacks; live supercomputing integration is an MoES-authorized post-hackathon deployment step.
- **IMD Onset Advisory Limitation:** IMD-referenced monsoon onset criteria adapted within the VarshaSetu hyperlocal forecasting framework; this dashboard does not issue an official statutory IMD onset declaration.
- **Rural Messaging Sandbox:** SMS and WhatsApp delivery workflows are demonstrated through a sandboxed simulation gateway without live commercial cellular telecom charges.
- **Cartographic Boundaries:** Demonstration village polygons and isochrones are approximate visualization bounding-boxes, not official Survey of India administrative cadastral boundaries.